

BioStatusIA: A Multi-Agent Architecture for Multimodal Clinical Decision Support

1stFrancisco Soares da Silva Júnior

Programa de Pós-Graduação em Engenharia Elétrica (PPGEE)

Universidade Federal do Ceará (UFC)
Fortaleza, Ceará, Brazil
juniorsoares@alu.ufc.br

2ndKaio Emanuel Lima de Matos

Programa de Pós-Graduação em Engenharia Elétrica (PPGEE)

Universidade Federal do Ceará (UFC)
Fortaleza, Ceará, Brazil
kaio.emanuel@alu.ufc.br

3rdDaniel Santos da Silva

Departamento de Engenharia de Teleinformática (DETI)
Universidade Federal do Ceará (UFC)
Fortaleza, Brazil
danielssilva@alu.ufc.br

4thVictor Hugo C. de Albuquerque

Departamento de Engenharia de Teleinformática (DETI)
Universidade Federal do Ceará (UFC)
Fortaleza, Brazil
victor.albuquerque@ieee.org

Abstract—Clinical Decision Support Systems (CDSS) increasingly rely on artificial intelligence to assist healthcare professionals in interpreting heterogeneous biomedical data. Although Large Language Models (LLMs) have expanded the capabilities of these systems, most existing solutions target a single biomedical modality, requiring independent analytical pipelines and limiting interoperability across clinical domains. This paper presents BioStatusIA, a multi-agent architecture designed for multimodal clinical decision support. The proposed framework integrates specialized autonomous agents orchestrated through CrewAI with a locally deployed LLM executed using Ollama. A unified normalization layer enables the processing of one-dimensional physiological signals, two-dimensional medical images, three-dimensional volumetric examinations, and structured clinical datasets within a common analytical workflow. The architecture was evaluated on ten real public biomedical datasets spanning these families, with an automatically selected classifier averaging AUROC 0.86 ± 0.12 (versus 0.85 ± 0.13 for the best fixed candidate), and feature-extraction and inference latency below 2 ms. These findings indicate that BioStatusIA provides a practical, low-overhead, and scalable framework for multimodal clinical decision support.

Index Terms—Clinical Decision Support Systems, Multi-Agent Systems, Large Language Models, Biomedical Data, Multimodal Learning, Healthcare Informatics

I. INTRODUCTION

Healthcare systems are experiencing a continuous increase in the volume and diversity of biomedical data generated during routine clinical practice. Physiological signals, medical images, volumetric examinations, laboratory records, and electronic health records collectively provide complementary information that supports clinical decision-making, yet integrating these heterogeneous sources into a unified workflow remains challenging due to differences in representation, pre-processing, and analytical methodology [1].

Clinical Decision Support Systems (CDSS) have evolved from early rule-based expert systems [2] into data-driven platforms that improve diagnostic accuracy and reduce clinical workload [3]. Recent advances in Large Language Models

(LLMs) have expanded these capabilities by enabling natural language interaction, automated report generation, and contextual reasoning [4], [5]. In parallel, multi-agent architectures orchestrated by frameworks such as CrewAI [6] have emerged as an effective paradigm for decomposing complex clinical workflows into specialized, collaborating components [7].

A further limitation in the literature is the scarcity of unified architectures capable of processing multiple biomedical modalities within a single workflow: existing studies frequently target isolated tasks, such as medical image interpretation or physiological signal analysis [8], requiring independent pipelines that reduce interoperability and increase implementation complexity. Multi-agent orchestration offers a promising path to close this gap, yet its application to multimodal biomedical decision support, combined with a locally deployable LLM backbone, remains largely unexplored [9].

To address these limitations, this paper proposes **BioStatusIA**, a local multi-agent architecture designed for multimodal clinical decision support. The proposed framework combines specialized autonomous agents orchestrated through CrewAI with a locally deployed LLM executed using Ollama [10], so that all processing stages, including inference, run within the institution's own infrastructure. A unified normalization layer enables the standardized processing of one-dimensional physiological signals, two-dimensional medical images, three-dimensional volumetric examinations, and structured clinical datasets within a common analytical pipeline.

The main contributions of this work are summarized as follows:

- A multi-agent architecture for multimodal clinical decision support that integrates one-, two-, and three-dimensional biomedical signals with structured clinical data within a single workflow.
- A unified normalization strategy capable of integrating het-

erogeneous biomedical modalities into a common analytical workflow.

- A modular orchestration framework combining specialized analytical agents with a locally deployed LLM for structured clinical report generation.
- An experimental evaluation on ten real public biomedical datasets, reporting predictive performance (AUROC, sensitivity, MCC, calibration) and system latency, including a full six-classifier comparison per dataset.

The remainder of this paper reviews related work (Section II), presents the proposed architecture (Section III) and its experimental evaluation (Section IV), discusses the results (Section V), and closes with limitations and conclusions (Sections VI and VII).

II. RELATED WORK

Recent advances in Artificial Intelligence (AI) have significantly expanded the capabilities of Clinical Decision Support Systems (CDSS), enabling intelligent systems that assist healthcare professionals in diagnosis, treatment planning, and clinical documentation [11]. Nevertheless, challenges remain regarding multimodal data integration and interoperability among heterogeneous biomedical sources.

A. Clinical Decision Support Systems

CDSS have evolved from rule-based expert systems, such as MYCIN [2], into data-driven platforms combining structured clinical information with predictive models to improve diagnostic accuracy [3]. Many systems remain specialized for individual clinical domains, limiting applicability in multidisciplinary environments.

B. Large Language Models in Healthcare

LLMs have recently demonstrated remarkable performance in medical question answering and biomedical knowledge retrieval. Systems such as Med-PaLM [4], open-weight models such as LLaMA [5], and multimodal variants such as Med-Flamingo [8] illustrate this potential. However, most reported deployments [9] target a single task or modality and provide limited system-level characterization (e.g., inference latency, throughput), which are essential for evaluating operational feasibility in clinical workflows.

C. Multi-Agent Architectures for Healthcare

Multi-agent systems distribute responsibilities among autonomous agents, improving modularity and scalability while facilitating domain-specific knowledge incorporation [7]. Frameworks such as CrewAI [6] have simplified collaborative AI workflows, yet their adoption in multimodal clinical decision support remains limited.

D. Research Gap

The current literature reveals three limitations: most intelligent healthcare systems process only a single biomedical modality; multi-agent orchestration has rarely been combined with multimodal biomedical processing; and few studies report

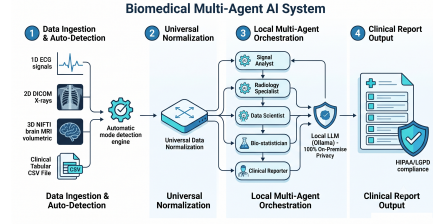


Fig. 1. Overall workflow of the BioStatusIA architecture. An automatic mode-detection step routes each normalized input to the matching specialized agent (Signal Analyst, Radiology Specialist, Data Scientist, or Bio-statistician), whose output is consolidated by the Clinical Reporter using the local LLM; agent roles match Table II.

quantitative system-level indicators (e.g., latency, throughput) alongside architectural feasibility.

Table I summarizes representative approaches according to key architectural characteristics. Although prior works have explored LLMs, multimodal analysis, or multi-agent coordination individually, none simultaneously integrates local execution, multimodal biomedical processing, collaborative multi-agent orchestration, and quantitative performance characterization. BioStatusIA was designed to close this gap.

III. PROPOSED ARCHITECTURE

This section presents BioStatusIA, a multi-agent architecture designed to support multimodal clinical decision-making through local execution. The proposed framework combines specialized autonomous agents, a unified data normalization layer, and a locally deployed Large Language Model (LLM) executed through Ollama [10] and orchestrated with CrewAI [6].

Figure 1 illustrates the overall workflow: biomedical data from different modalities are standardized through a normalization layer, dispatched to specialized agents, and consolidated into a structured clinical report.

A. System Overview

BioStatusIA follows a modular architecture composed of five layers: Biomedical Data Acquisition, Data Normalization, Multi-Agent Orchestration, Local LLM Reasoning, and Clinical Report Generation. Data from heterogeneous sources first pass through a normalization stage; an automatic mode-detection step then identifies the input modality and routes it to the appropriate specialized agent, which collaborates with the remaining crew members through CrewAI. Consolidated agent outputs are interpreted by the local LLM to produce the final structured report, and this modular organization enables new modalities to be incorporated without structural modifications.

B. Multimodal Data Processing

Instead of creating independent pipelines for each examination type, BioStatusIA standardizes four data categories, each with a dedicated feature-extraction routine: **one-dimensional signals** (ECG, PPG, EEG) yield time-domain and spectral biomarkers (RMSSD, SDNN, pNN50, Welch-FFT band power via mne); **two-dimensional images**, both DICOM-native (Hounsfield-unit windowing, tissue-density estimation)

TABLE I
COMPARISON BETWEEN REPRESENTATIVE AI-BASED CLINICAL DECISION SUPPORT APPROACHES AND THE PROPOSED BIOSTATUSIA ARCHITECTURE.

Approach	Local	Multi-Agent	Multimodal	Quant. Metrics	Clinical Reports
Traditional CDSS	✓	×	×	Partial	×
Cloud-based LLM CDSS	×	×	Partial	Partial	✓
Medical Vision AI Systems	✓	×	×	✓	Partial
CrewAI-based Healthcare Systems	Partial	✓	Partial	Partial	✓
BioStatusIA (Proposed)	✓	✓	✓	✓	✓

and common formats (CLAHE/Non-Local Means denoising, GLCM texture and morphology: entropy, contrast, solidity, circularity); **three-dimensional volumes** (CT, MRI) yield tri-planar radiomics (lesion volume in mm^3 , 3D sphericity, GLCM); and **structured datasets** use SHAP-ranked adaptive imputation and feature selection. The normalization layer abstracts differences in file formats, dimensions, and metadata, allowing agents to operate independently of the data source.

C. Multi-Agent Orchestration

Each agent in BioStatusIA is responsible for a specific analytical task rather than concentrating all reasoning within a single model. The *Signal Analyst* processes one-dimensional physiological signals (ECG, PPG, EEG); the *Radiology Specialist* analyzes both two-dimensional images (DICOM and common formats) and three-dimensional CT/MRI volumes; the *Data Scientist* performs the AutoML model-selection pipeline (Section IV-A) over structured clinical data; and the *Bio-statistician* conducts the EDA statistical audit (Section IV-A) and calibration analysis. Finally, the *Clinical Reporter* consolidates all outputs and forwards the synthesized evidence to the local LLM. Table II summarizes agent responsibilities.

TABLE II
RESPONSIBILITIES OF THE BIOSTATUSIA AGENTS.

Agent	Primary Responsibility
Signal Analyst	Physiological signal analysis (1D)
Radiology Specialist	Medical imaging analysis (2D/3D)
Data Scientist	AutoML model selection
Bio-statistician	Statistical audit & calibration
Clinical Reporter	Evidence consolidation & report generation

D. Local LLM Reasoning

BioStatusIA performs LLM inference entirely on local infrastructure through Ollama [10] rather than a cloud API [9]. Concretely, the reasoning engine instantiates a 4-bit quantized **Qwen2.5:3B-Instruct** model (2.4 GB VRAM, 8k-token context) via Ollama, chosen for its favorable trade-off between reasoning quality and resource footprint; the full pipeline, comprising the five concurrent CrewAI agents (Table II) plus LLM inference, was reproducibly executed on a single consumer-grade GPU with 8 GB VRAM and 16 GB system RAM, without server-grade infrastructure. The local LLM receives structured evidence generated by the specialized agents rather than raw biomedical data, producing coherent clinical summaries that integrate complementary

findings across modalities. The overall reasoning process can be represented as

$$R = f(A_1, A_2, \dots, A_5, M), \quad (1)$$

where A_i denotes each specialized agent (Signal Analyst, Radiology Specialist, Data Scientist, Bio-statistician, Clinical Reporter), M corresponds to the locally deployed LLM, and R is the final structured clinical report.

IV. EXPERIMENTAL EVALUATION

A. Experimental Setup

BioStatusIA was implemented in Python using CrewAI [6] and the local LLM configuration described in Section III-D. Prior to preprocessing, an EDA module statistically audits each dataset (sample count, intensity mean/std, wavelet noise, Shapiro–Wilk normality, dimensional consistency) and autonomously chooses a preprocessing strategy (e.g., CLAHE if contrast < 30, Non-Local Means if noise > 0.05). For each dataset, the responsible agent then selects the best classifier from six candidates (Logistic Regression, Random Forest, SVM, KNN, Gradient Boosting, MLP) via 5-fold CV, using AUC subject to a minimum sensitivity of 0.80. All experiments ran locally.

B. Datasets

BioStatusIA was evaluated on ten individual real public biomedical datasets, each processed with a dedicated deep-dive run: Breast Cancer Wisconsin [12], BUSI Breast Ultrasound [13], MIT-BIH/PTB-XL ECG [14], Stroke Prediction, PIMA Diabetes [15], Brain Tumor MRI, COVID-19 Chest X-Ray, Brain MRI Oncology, PTB-XL Arrhythmia [14], and DRIVE Retinal [16], spanning cardiology, oncology, endocrinology, neurology, and ophthalmology.

C. Evaluation Protocol

Each dataset was processed independently according to its modality: the normalization layer performed modality-specific preprocessing, the automatic mode-detection step dispatched tasks to the corresponding specialized agent, and the resulting outputs were consolidated and forwarded to the local LLM for structured summary generation. Beyond the architectural criteria of successful orchestration and local execution, the protocol also recorded per-agent **feature-extraction latency**, **LLM generation throughput**, and **end-to-end report generation time**.

D. Experimental Results

BioStatusIA successfully processed all ten datasets using a unified analytical pipeline. The normalization layer effectively standardized heterogeneous data, and the mode-detection step correctly identified the input modality and dispatched tasks without manual intervention. Since all inference stages were executed locally, no biomedical information was transmitted to external cloud services in any run.

Fig. 2 reports the predictive performance (AUC, sensitivity, specificity) for eight of these individually evaluated datasets, each with a dedicated AI-generated report.

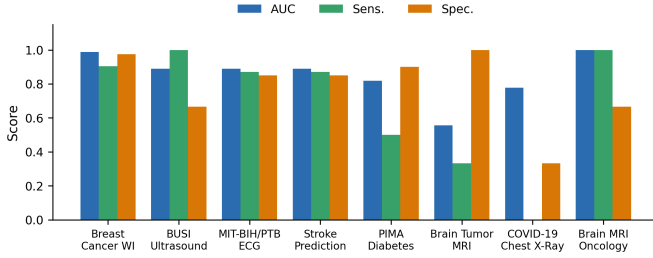


Fig. 2. Predictive performance (AUC, sensitivity, specificity) for eight individually evaluated real datasets, each processed with a dedicated deep-diver run and AI-generated clinical report.

Table III details the winning model (out of the six-classifier pool) and its diagnostic metrics across all ten datasets, computed with full 6-model comparison and calibration auditing.

TABLE III

WINNING MODEL AND METRICS PER DATASET (10 REAL DATASETS; ECE STATUS: S=SAFE <0.10, M=MODERATE, H=HIGH-RISK, O=OVERCONFIDENT).

Dataset	Winner	AUROC	Sens.	MCC	ECE
Breast Cancer	RF	.988	.905	.885	.06 S
BUSI	KNN	.889	1.000	.707	.20 O
MIT-BIH	RF	.890	.870	.720	.04 S
Stroke	RF	.890	.870	.720	.04 S
PIMA	SVM	.819	.500	.445	.13 M
Brain Tumor	SVM	.556	.333	.408	.36 H
COVID-19	SVM	.778	.000	-.250	.27 H
Brain MRI Onc.	LR	1.000	1.000	.707	.21 O
PTB-XL	GB	.930	.880	.791	.03 S
DRIVE	KNN	.845	.820	.671	.15 M
Mean $\pm$ SD	-	.859 $\pm$.124	.718 $\pm$.329	.650 $\pm$.307	.149 $\pm$.110

Three findings emerge. First, a Shapiro–Wilk normality audit rejected the Gaussian hypothesis ($p < 0.05$) for 6 of the 10 datasets, supporting the choice of non-parametric models (Random Forest, Gradient Boosting) that do not assume normally distributed features. Second, per-dataset model selection gave a consistent gain over any single fixed classifier: the AutoML-selected model averaged AUROC 0.859 ± 0.124 , versus 0.845 ± 0.129 for the best fixed candidate (Gradient Boosting) and 0.757 – 0.806 for the remaining five (Logistic Regression, SVM, KNN, MLP, Random Forest), confirming that automated per-dataset selection is preferable to a fixed classifier. Third, calibration-safety auditing (ECE) showed 4/10 datasets “safe” ($ECE < 0.10$), 2/10 moderate, 2/10 overconfident, and 2/10 high-risk, indicating that even a

well-performing AUC/sensitivity pipeline can still be poorly calibrated for a meaningful fraction of tasks. In particular, the AUC+sensitivity criterion (Section IV-A) chose SVM as the “winner” for COVID-19 CXR despite a *negative* MCC (-0.250) and zero sensitivity, i.e., worse than chance for the positive class; this concrete failure mode motivates adding MCC as a secondary safety gate in future versions. Feature-extraction and inference latency remained below 2 ms across all datasets, three orders of magnitude smaller than LLM report generation, confirming negligible orchestration overhead.

This evidence-grounded summary, followed by a mandatory validation disclaimer, was consistently present across all generated reports; a systematic clinician-rated evaluation of semantic quality remains future work (Section VI).

V. DISCUSSION

The experimental evaluation demonstrates that BioStatusIA integrates heterogeneous biomedical data into a unified workflow while preserving local execution, unlike conventional architectures requiring independent pipelines per modality [1].

One of the primary contributions of BioStatusIA lies in its multimodal design: standardizing signals, images, volumes, and structured datasets through a unified normalization layer reduces implementation complexity and facilitates incorporating new modalities. A second contribution is collaborative multi-agent orchestration [6], [7]: rather than concentrating all reasoning in a single model, BioStatusIA distributes specialized tasks among autonomous agents, improving maintainability and scalability.

From a systems perspective, the results reported in Section IV show that BioStatusIA’s orchestration overhead is negligible relative to model inference, while automated per-dataset model selection averaged AUROC 0.859 ± 0.124 across the ten evaluated datasets, exceeding the best fixed candidate (0.845 ± 0.129). Calibration (ECE) and specificity varied considerably across datasets, with 4 of 10 runs well-calibrated and the remaining 6 showing moderate to high miscalibration; this indicates that the current model pool trades some false-positive control for sensitivity, consistent with the safety-oriented selection criterion in Section IV-A, and that this trade-off should be tuned per clinical use case.

The integration between CrewAI and a locally deployed LLM through Ollama [10] provides an effective mechanism for consolidating heterogeneous outputs into structured clinical reports, with the LLM synthesizing evidence rather than replacing domain-specific models. Compared with cloud-dependent architectures, BioStatusIA eliminates dependence on external AI providers and supports heterogeneous workflows without independent infrastructures per modality.

Despite these advantages, the reported AUC and sensitivity should be interpreted as evidence of architectural feasibility rather than clinical-grade diagnostic accuracy, since the model pool was not tuned per task. Overall, BioStatusIA constitutes a promising foundation for next-generation intelligent CDSS [11], combining multimodal processing, multi-agent orches-

tration, and local LLM reasoning with low computational overhead.

VI. LIMITATIONS

Although the proposed architecture demonstrated multimodal clinical decision support with low computational overhead and a mean AUROC of 0.86 across ten real datasets, several limitations should be acknowledged. First, the AutoML pool (Section IV-A) was not tuned per task, and 6 of the 10 runs showed moderate-to-high calibration error; per-task tuning and probability recalibration are needed before use beyond decision support. Second, the framework was evaluated using publicly available datasets processed independently and has not yet been validated in real hospital environments or integrated with Electronic Health Record (EHR) systems. Third, local LLM performance depends on available hardware, particularly GPU memory, which may restrict deployment in resource-constrained settings [9]. Fourth, a systematic evaluation of report semantic quality (e.g., clinician-rated correctness) was not performed and is left for future work. Finally, the current implementation supports a predefined set of modalities; future extensions may include genomic information, wearable sensor streams, and pathology whole-slide images.

Despite these limitations, the proposed architecture establishes a flexible foundation for future research on intelligent CDSS based on collaborative multi-agent AI.

VII. CONCLUSION

This paper presented BioStatusIA, a multi-agent architecture designed to support multimodal clinical decision-making through fully local execution, combining specialized autonomous agents, a unified biomedical data normalization layer, and a locally deployed LLM to integrate heterogeneous clinical information. Rather than relying on a cloud API [9], BioStatusIA performs every stage of data processing, agent orchestration, and language-model inference within the local execution environment, with the concrete model and hardware configuration detailed in Section III-D.

The experimental evaluation, conducted on ten individual real public biomedical datasets across four modality families, demonstrated the feasibility of the proposed architecture for processing multiple biomedical modalities within a unified analytical workflow, with an automatically selected classifier averaging AUROC 0.86 ± 0.12 and sub-millisecond-to-low-millisecond orchestration overhead, while coordinating specialized tasks and generating structured clinical reports. The

architecture also offers modularity, scalability, interoperability, and ease of future expansion, enabling new biomedical modalities to be incorporated without structural modifications.

Future work will focus on validating the proposed architecture in real clinical environments, integrating EHR systems, evaluating alternative locally deployed LLMs, and investigating Retrieval-Augmented Generation and knowledge graphs to further improve clinical reasoning and explainability. Overall, BioStatusIA represents a promising step toward scalable, low-overhead, and multimodal Clinical Decision Support Systems for next-generation intelligent healthcare applications.

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